

# Forecasting in Macroeconomics

Martynas Baltramaitis  
Applied Macroeconomics

November 5, 2025

In this document I will provide answers to the assignment questions, graphs and other objects. The code used I will provide as a .R file when submitting my answers.

## 1 Question

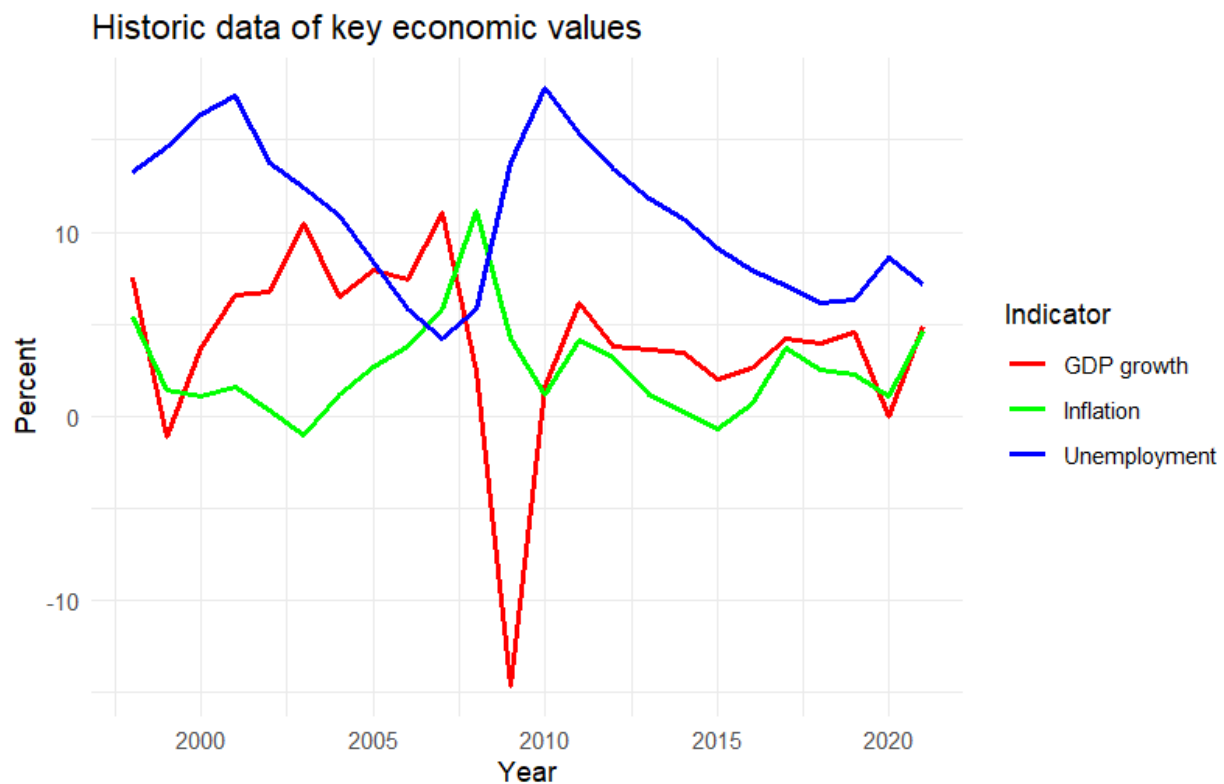


Figure 1: Historic data of key economic values

The past dynamics of these variables do help predicts the future movements of the economy as we can observe business cycles in variables like plotted out. Just in this graph we can

observe GDP growth moving in cycles and employment inversely responding to the changes in GDP growth. Similarly we can observe how inflation reacts to GDP growth as another example.

In this case I would say it is better to try to use multivariate information set to try to leverage all three time series simultaneously. This can help better explain variables that have interconnecting dependencies. For example if GDP growth impacts unemployment, perhaps we can better estimate future unemployment if we use GDP growth in our prediction. I understand that this approach does not quite follow the parsimony principle and it might be difficult to fully implement due to having not too large of a data set.

## 2 Question

Here are my forecasts:

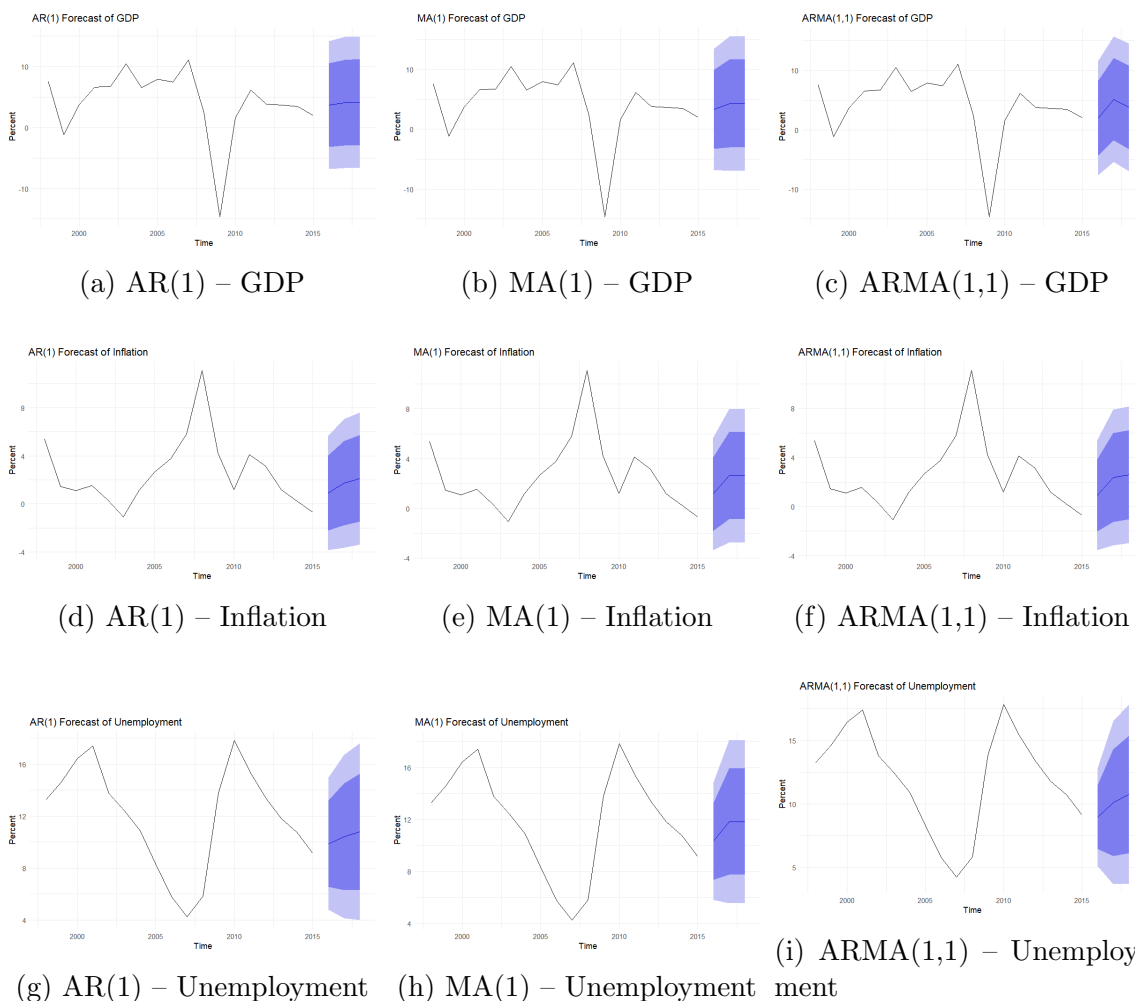


Figure 2: Forecast Comparison Across Models and Variables (in sample 1998 to 2015)

I also have made a table which contains the errors saved:

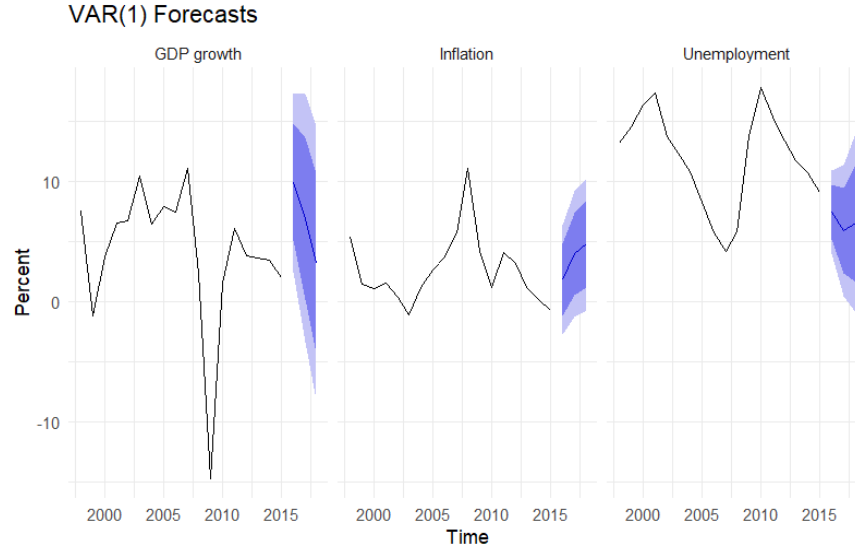


Figure 3: VAR(1) Model Forecasts (in sample 1998 to 2015)

Table 1: Forecast Errors by Model, Variable and Forecast Horizon (in sample 1998 to 2015)

| Model     | Variable     | Horizon | Error      |
|-----------|--------------|---------|------------|
| AR(1)     | GDP          | 1       | -1.0455249 |
| AR(1)     | GDP          | 2       | 0.0861194  |
| AR(1)     | GDP          | 3       | -0.2055951 |
| MA(1)     | GDP          | 1       | -0.6623952 |
| MA(1)     | GDP          | 2       | -0.1061353 |
| MA(1)     | GDP          | 3       | -0.3064593 |
| ARMA(1,1) | GDP          | 1       | 0.6503786  |
| ARMA(1,1) | GDP          | 2       | -0.9874121 |
| ARMA(1,1) | GDP          | 3       | 0.1789538  |
| AR(1)     | Inflation    | 1       | -0.2326574 |
| AR(1)     | Inflation    | 2       | 1.9939372  |
| AR(1)     | Inflation    | 3       | 0.3978293  |
| MA(1)     | Inflation    | 1       | -0.4729166 |
| MA(1)     | Inflation    | 2       | 1.0926601  |
| MA(1)     | Inflation    | 3       | -0.0889701 |
| ARMA(1,1) | Inflation    | 1       | -0.2514184 |
| ARMA(1,1) | Inflation    | 2       | 1.3435939  |
| ARMA(1,1) | Inflation    | 3       | -0.0552075 |
| AR(1)     | Unemployment | 1       | -1.9668223 |
| AR(1)     | Unemployment | 2       | -3.3085530 |
| AR(1)     | Unemployment | 3       | -4.6158032 |
| MA(1)     | Unemployment | 1       | -2.4157402 |
| MA(1)     | Unemployment | 2       | -4.7157398 |
| MA(1)     | Unemployment | 3       | -5.6394578 |
| ARMA(1,1) | Unemployment | 1       | -1.0496732 |
| ARMA(1,1) | Unemployment | 2       | -2.9855217 |
| ARMA(1,1) | Unemployment | 3       | -4.5580952 |
| VAR(1)    | GDP          | 1       | -7.3163500 |
| VAR(1)    | GDP          | 2       | -2.7834040 |
| VAR(1)    | GDP          | 3       | 0.7383446  |
| VAR(1)    | Inflation    | 1       | -1.1588745 |
| VAR(1)    | Inflation    | 2       | -0.2825598 |
| VAR(1)    | Inflation    | 3       | -2.3019302 |
| VAR(1)    | Unemployment | 1       | 0.4118703  |
| VAR(1)    | Unemployment | 2       | 1.1687058  |
| VAR(1)    | Unemployment | 3       | -0.3603115 |

### 3 Question

Repeating the exercise two more times I provide the error tables and the forecast figures.

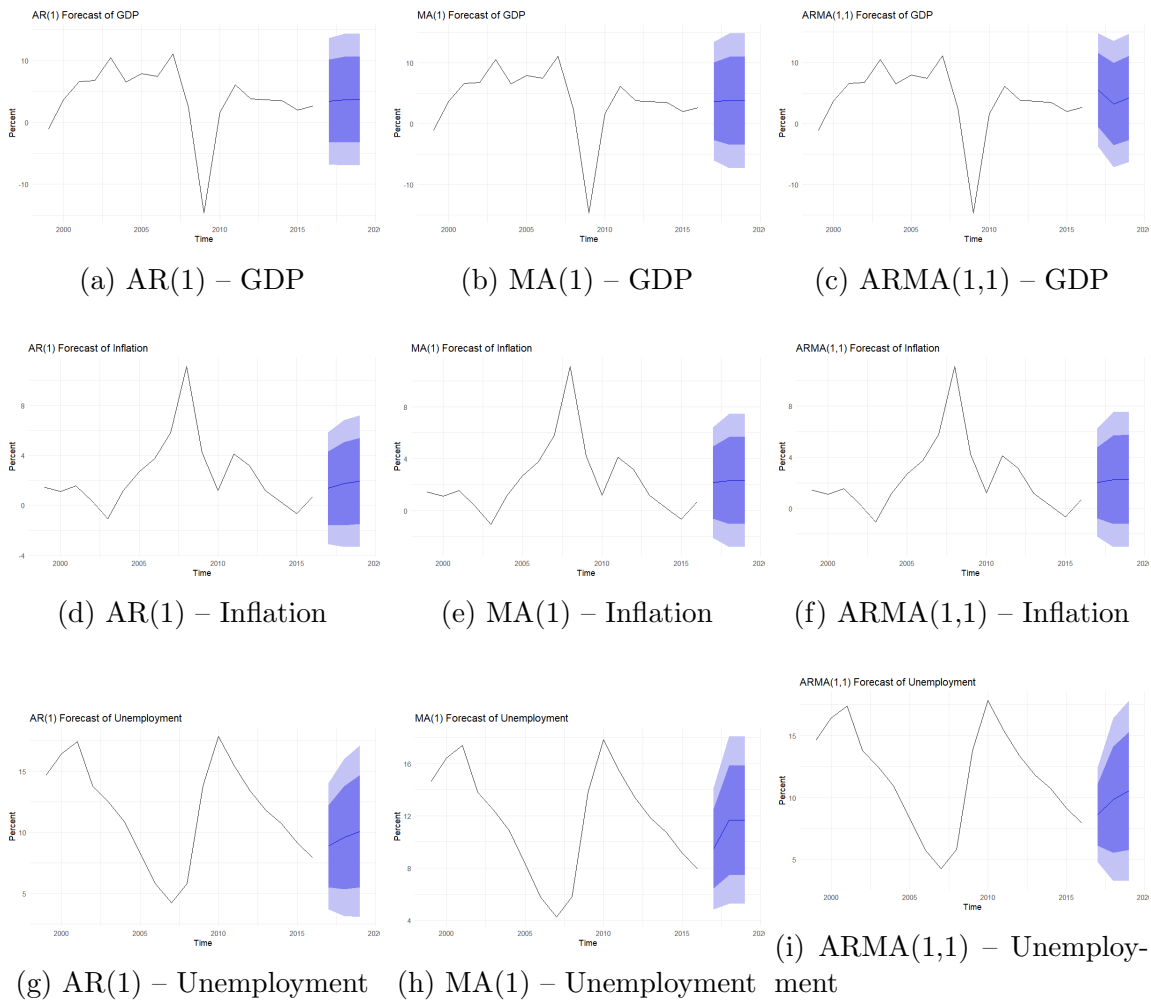


Figure 4: Forecast Comparison Across Models and Variables (in sample 1999 to 2016)

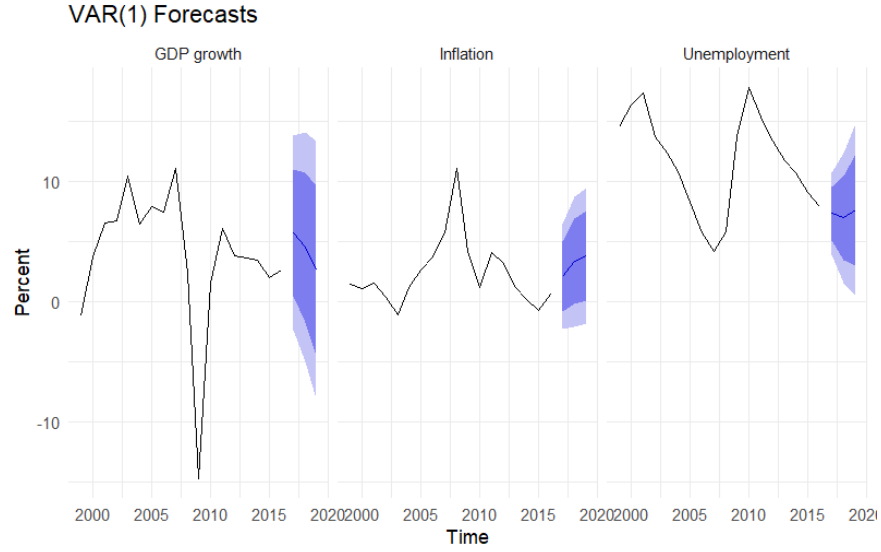


Figure 5: VAR(1) Model Forecasts (in sample 1999 to 2016)

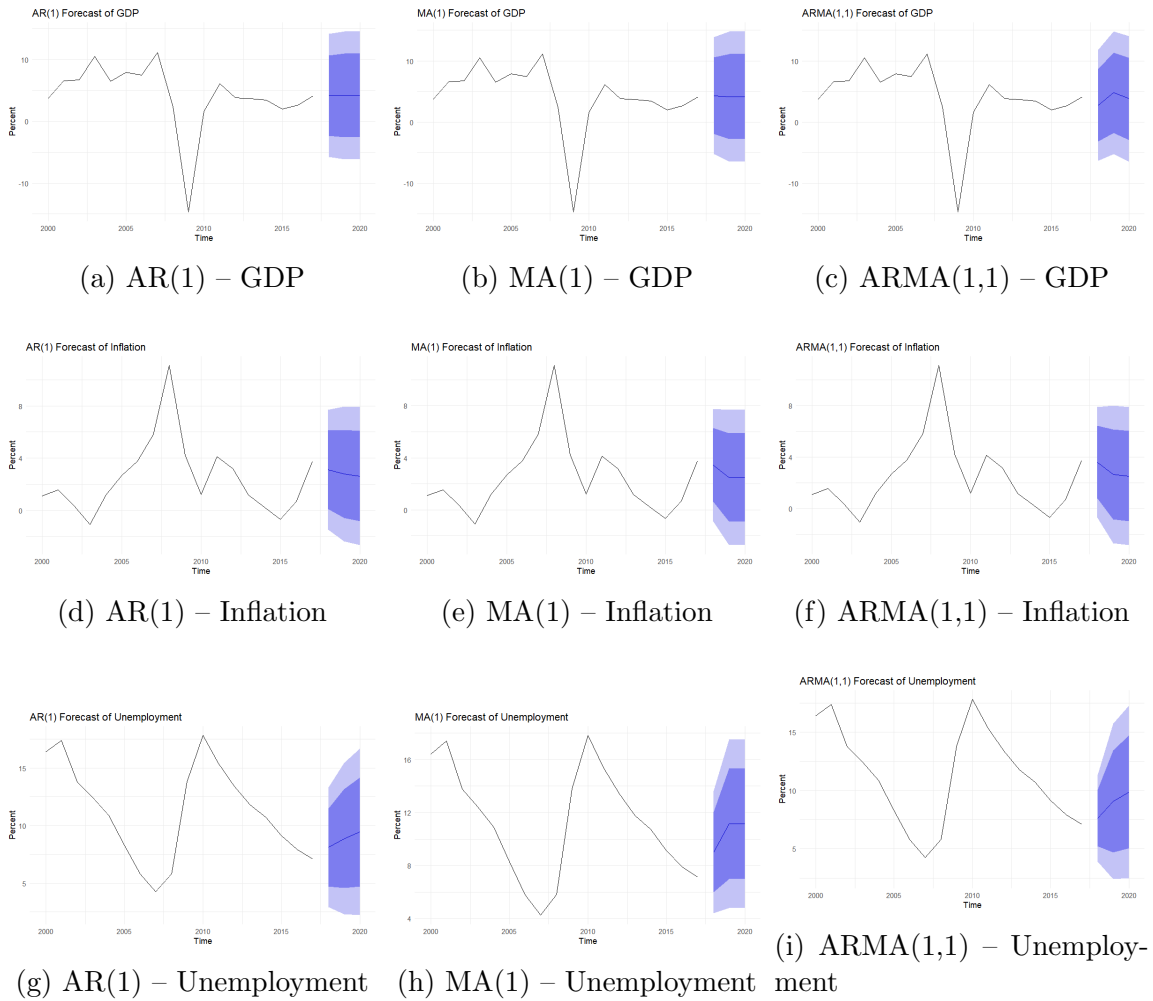


Figure 6: Forecast Comparison Across Models and Variables (in sample 2000 to 2017)

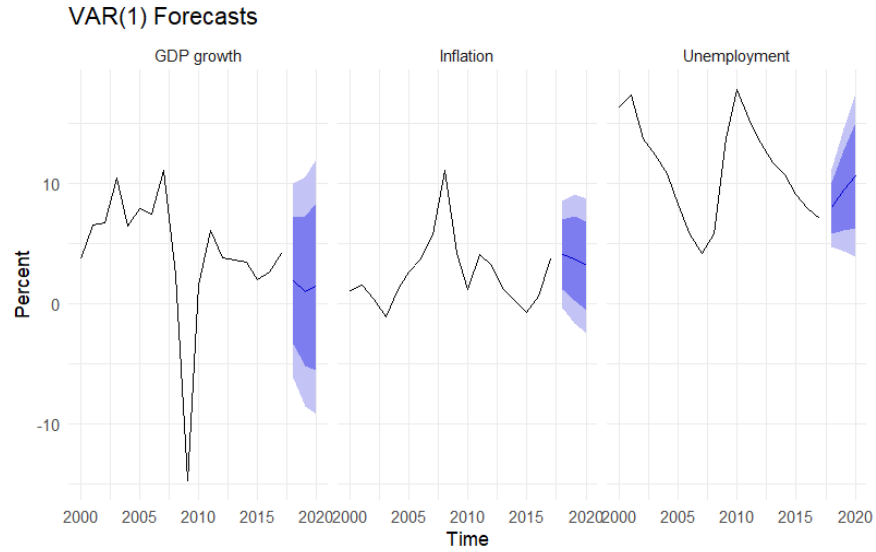


Figure 7: VAR(1) Model Forecasts (in sample 2000 to 2017)

Table 2: Forecast Errors by Model, Variable and Forecast Horizon (in sample 1999 to 2016)

| Model     | Variable     | Horizon | Error      |
|-----------|--------------|---------|------------|
| AR(1)     | GDP          | 1       | 0.7132426  |
| AR(1)     | GDP          | 2       | 0.2827262  |
| AR(1)     | GDP          | 3       | 0.8295011  |
| MA(1)     | GDP          | 1       | 0.4996243  |
| MA(1)     | GDP          | 2       | 0.1843784  |
| MA(1)     | GDP          | 3       | 0.7965430  |
| ARMA(1,1) | GDP          | 1       | -1.3291272 |
| ARMA(1,1) | GDP          | 2       | 0.7444924  |
| ARMA(1,1) | GDP          | 3       | 0.3633662  |
| AR(1)     | Inflation    | 1       | 2.3491625  |
| AR(1)     | Inflation    | 2       | 0.7936462  |
| AR(1)     | Inflation    | 3       | 0.2985767  |
| MA(1)     | Inflation    | 1       | 1.5614843  |
| MA(1)     | Inflation    | 2       | 0.2258345  |
| MA(1)     | Inflation    | 3       | -0.0663433 |
| ARMA(1,1) | Inflation    | 1       | 1.7122139  |
| ARMA(1,1) | Inflation    | 2       | 0.2899577  |
| ARMA(1,1) | Inflation    | 3       | -0.0439827 |
| AR(1)     | Unemployment | 1       | -1.7565397 |
| AR(1)     | Unemployment | 2       | -3.3817360 |
| AR(1)     | Unemployment | 3       | -3.7986350 |
| MA(1)     | Unemployment | 1       | -2.3833325 |
| MA(1)     | Unemployment | 2       | -5.4812163 |
| MA(1)     | Unemployment | 3       | -5.3791650 |
| ARMA(1,1) | Unemployment | 1       | -1.5212685 |
| ARMA(1,1) | Unemployment | 2       | -3.6440095 |
| ARMA(1,1) | Unemployment | 3       | -4.2624302 |
| VAR(1)    | GDP          | 1       | -1.5755996 |
| VAR(1)    | GDP          | 2       | -0.5899353 |
| VAR(1)    | GDP          | 3       | 1.9369455  |
| VAR(1)    | Inflation    | 1       | 1.6019520  |
| VAR(1)    | Inflation    | 2       | -0.8360646 |
| VAR(1)    | Inflation    | 3       | -1.6141519 |
| VAR(1)    | Unemployment | 1       | -0.2450904 |
| VAR(1)    | Unemployment | 2       | -0.8051173 |
| VAR(1)    | Unemployment | 3       | -1.3897975 |

Table 3: Forecast Errors by Model, Variable and Forecast Horizon (in sample 2000 to 2017)

| Model     | Variable     | Horizon | Error      |
|-----------|--------------|---------|------------|
| AR(1)     | GDP          | 1       | -0.2189278 |
| AR(1)     | GDP          | 2       | 0.3880551  |
| AR(1)     | GDP          | 3       | -4.1980233 |
| MA(1)     | GDP          | 1       | -0.3362750 |
| MA(1)     | GDP          | 2       | 0.4135721  |
| MA(1)     | GDP          | 3       | -4.1710631 |
| ARMA(1,1) | GDP          | 1       | 1.2151456  |
| ARMA(1,1) | GDP          | 2       | -0.2094340 |
| ARMA(1,1) | GDP          | 3       | -3.7988494 |
| AR(1)     | Inflation    | 1       | -0.5651042 |
| AR(1)     | Inflation    | 2       | -0.5416205 |
| AR(1)     | Inflation    | 3       | -1.5535967 |
| MA(1)     | Inflation    | 1       | -0.9225693 |
| MA(1)     | Inflation    | 2       | -0.2419659 |
| MA(1)     | Inflation    | 3       | -1.4155608 |
| ARMA(1,1) | Inflation    | 1       | -1.0619436 |
| ARMA(1,1) | Inflation    | 2       | -0.3915193 |
| ARMA(1,1) | Inflation    | 3       | -1.4367275 |
| AR(1)     | Unemployment | 1       | -1.9178788 |
| AR(1)     | Unemployment | 2       | -2.5784643 |
| AR(1)     | Unemployment | 3       | -0.8809310 |
| MA(1)     | Unemployment | 1       | -2.8292722 |
| MA(1)     | Unemployment | 2       | -4.8722197 |
| MA(1)     | Unemployment | 3       | -2.5896556 |
| ARMA(1,1) | Unemployment | 1       | -1.4597812 |
| ARMA(1,1) | Unemployment | 2       | -2.7882947 |
| ARMA(1,1) | Unemployment | 3       | -1.3251867 |
| VAR(1)    | GDP          | 1       | 2.0261689  |
| VAR(1)    | GDP          | 2       | 3.5487473  |
| VAR(1)    | GDP          | 3       | -1.4582210 |
| VAR(1)    | Inflation    | 1       | -1.5979960 |
| VAR(1)    | Inflation    | 2       | -1.5479821 |
| VAR(1)    | Inflation    | 3       | -2.1014507 |
| VAR(1)    | Unemployment | 1       | -1.8072169 |
| VAR(1)    | Unemployment | 2       | -3.1735507 |
| VAR(1)    | Unemployment | 3       | -2.1331967 |

## 4 Question

To comment on the model prediction accuracy I made the following table to compare the results.

Table 4: Forecasting model accuracy measure table

| InSample  | Model     | Variable     | MSE    | MSPE       | RMSE  | RMSPE   | MAE   | RMAPE   |
|-----------|-----------|--------------|--------|------------|-------|---------|-------|---------|
| 1998-2015 | AR(1)     | GDP          | 0.381  | 0.053      | 0.617 | 0.230   | 0.446 | 0.156   |
| 1998-2015 | AR(1)     | Inflation    | 1.396  | 0.143      | 1.182 | 0.379   | 0.875 | 0.345   |
| 1998-2015 | AR(1)     | Unemployment | 12.040 | 0.278      | 3.470 | 0.528   | 3.297 | 0.487   |
| 1998-2015 | ARMA(1,1) | GDP          | 0.477  | 0.040      | 0.690 | 0.199   | 0.606 | 0.176   |
| 1998-2015 | ARMA(1,1) | Inflation    | 0.624  | 0.089      | 0.790 | 0.299   | 0.550 | 0.251   |
| 1998-2015 | ARMA(1,1) | Unemployment | 10.264 | 0.246      | 3.204 | 0.496   | 2.864 | 0.430   |
| 1998-2015 | MA(1)     | GDP          | 0.181  | 0.023      | 0.426 | 0.152   | 0.358 | 0.118   |
| 1998-2015 | MA(1)     | Inflation    | 0.475  | 0.191      | 0.689 | 0.437   | 0.552 | 0.342   |
| 1998-2015 | MA(1)     | Unemployment | 19.959 | 0.455      | 4.468 | 0.674   | 4.257 | 0.627   |
| 1998-2015 | VAR(1)    | GDP          | 20.607 | 2.710      | 4.540 | 1.646   | 3.613 | 1.207   |
| 1998-2015 | VAR(1)    | Inflation    | 2.241  | 1.248      | 1.497 | 1.117   | 1.248 | 0.897   |
| 1998-2015 | VAR(1)    | Unemployment | 0.555  | 0.011      | 0.745 | 0.105   | 0.647 | 0.092   |
| 1999-2016 | AR(1)     | GDP          | 0.426  | 0.022      | 0.652 | 0.150   | 0.608 | 0.141   |
| 1999-2016 | AR(1)     | Inflation    | 2.079  | 0.172      | 1.442 | 0.414   | 1.147 | 0.359   |
| 1999-2016 | AR(1)     | Unemployment | 9.650  | 0.242      | 3.107 | 0.492   | 2.979 | 0.466   |
| 1999-2016 | ARMA(1,1) | GDP          | 0.818  | 0.048      | 0.904 | 0.218   | 0.812 | 0.195   |
| 1999-2016 | ARMA(1,1) | Inflation    | 1.006  | 0.075      | 1.003 | 0.274   | 0.682 | 0.198   |
| 1999-2016 | ARMA(1,1) | Unemployment | 11.254 | 0.284      | 3.355 | 0.533   | 3.143 | 0.494   |
| 1999-2016 | MA(1)     | GDP          | 0.306  | 0.016      | 0.553 | 0.125   | 0.494 | 0.113   |
| 1999-2016 | MA(1)     | Inflation    | 0.831  | 0.062      | 0.912 | 0.249   | 0.618 | 0.180   |
| 1999-2016 | MA(1)     | Unemployment | 21.553 | 0.543      | 4.643 | 0.737   | 4.415 | 0.692   |
| 1999-2016 | VAR(1)    | GDP          | 2.194  | 0.115      | 1.481 | 0.339   | 1.367 | 0.316   |
| 1999-2016 | VAR(1)    | Inflation    | 1.957  | 0.271      | 1.399 | 0.520   | 1.351 | 0.493   |
| 1999-2016 | VAR(1)    | Unemployment | 0.880  | 0.022      | 0.938 | 0.149   | 0.813 | 0.129   |
| 2000-2017 | AR(1)     | GDP          | 5.941  | 353331.956 | 2.437 | 594.417 | 1.602 | 343.234 |
| 2000-2017 | AR(1)     | Inflation    | 1.009  | 0.739      | 1.004 | 0.860   | 0.887 | 0.639   |
| 2000-2017 | AR(1)     | Unemployment | 3.701  | 0.092      | 1.924 | 0.303   | 1.792 | 0.274   |
| 2000-2017 | ARMA(1,1) | GDP          | 5.317  | 289332.638 | 2.306 | 537.896 | 1.741 | 310.672 |
| 2000-2017 | ARMA(1,1) | Inflation    | 1.115  | 0.670      | 1.056 | 0.819   | 0.963 | 0.646   |
| 2000-2017 | ARMA(1,1) | Unemployment | 3.887  | 0.092      | 1.972 | 0.304   | 1.858 | 0.278   |
| 2000-2017 | MA(1)     | GDP          | 5.894  | 348808.252 | 2.428 | 590.600 | 1.640 | 341.041 |
| 2000-2017 | MA(1)     | Inflation    | 0.971  | 0.632      | 0.985 | 0.795   | 0.860 | 0.598   |
| 2000-2017 | MA(1)     | Unemployment | 12.817 | 0.300      | 3.580 | 0.548   | 3.430 | 0.512   |
| 2000-2017 | VAR(1)    | GDP          | 6.275  | 42632.699  | 2.505 | 206.477 | 2.344 | 119.638 |
| 2000-2017 | VAR(1)    | Inflation    | 3.122  | 1.578      | 1.767 | 1.256   | 1.749 | 1.095   |
| 2000-2017 | VAR(1)    | Unemployment | 5.963  | 0.134      | 2.442 | 0.366   | 2.371 | 0.349   |

To begin with, I would like to address the elephant in the table of huge MSPE errors for



In Sample period of 2000-2017. This error comes mainly from the COVID shock and the forecast is just completely off the mark in that case.

Talking about which models produce the most accurate estimates, for GDP growth MSE suggests that MA(1) is the best approach for two of the three periods which we estimate. MSPE suggests the same thing. It only differs for the last period, where MSE suggests ARMA(1,1) to be strongest and MSPE suggesting that the VAR(1) model is the most accurate. Having this in mind, MA(1) likely best represents Lithuanian GDP growth dynamics.

For Inflation, MSE suggests the best forecasting model is MA(1) process and it is so in all of the periods. The MSPE would suggest that ARMA(1,1) performs best in the first period, MA(1) in the second period as well as in the third. In this case I would conclude that the MA(1) is the best forecasting model for inflation in Lithuania based on these results.

There is more of a difference when looking at Unemployment. Here VAR(1) model performs best according to MSE in the first two periods, while the AR(1) takes the crown for the final period. The same happens if we consider MSPE. For unemployment then, I would consider VAR(1) model the best at forecasting it.

My results do not confirm that using multivariate approach is better than univariate ones for GDP growth and Inflation. However, the multivariate model is better than univariate ones at forecasting Unemployment.

## 5 Question

For GDP Growth I pick: MA(1) and VAR(1) model since for RMSE it is the best in the third period (tie-break). For Inflation: MA(1) and ARMA(1,1). For Unemployment: VAR(1,1) and AR(1).

None of the Diebold-Mariano tests show that the difference between errors made by forecasts are statistically significant for all samples. That would suggest that there is not really much difference in how two closest "competitors" perform. So for each pair, each model could be valid choices. But I would also be hesitant to state that this result is true because we do not have too much data to perform the tests.